Pasang Bhote

929-867-9924 | plb6363@nyu.edu | pasangb.github.io | linkedin.com/in/pasang-bhote

EDUCATION

New York University, Tandon School of Engineering — May 2028
B.S. in Electrical and Computer Engineering — GPA: 3.7
Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Circuits, Digital Logic, Electronics, Computer Architecture and Organization, Embedded Systems

Projects

Hardware Team Member, *ARC Robotics (design robots for competition)* - New York, NY — Sep 2025 - Present

- Design the schematic and PCB layout in Altium Designer for an LM74900 and N-channel MOSFET test board to evaluate switching for a planned supercapacitor power-boost system.
- Designed a 24V to 12V DC to DC buck converter PCB to supply an NVIDIA Jetson.

Hardware Designer, *Smart Solar-Powered Vehicle* - New York, NY — Summer 2026

- Built an ESP32-based, smartphone-controlled vehicle with ultrasonic obstacle detection and rain sensing.
- Integrated a solar charging system, motor driver, sensors, and dual-motor front-wheel-drive system.
- Designed CAD components and performed system wiring, soldering, assembly, and hardware testing.

Designer and Programmer, *Posture Monitoring Device* - New York, NY — Jan - May 2025

- Designed a Raspberry Pi-based posture monitoring system using computer vision and GPIO-controlled buzzer feedback.
- Created circuit schematics in Fritzing and soldered custom circuitry for prototyping.
- Designed and fabricated a custom enclosure using Fusion 360 and Epilog laser cutting.

Special Projects Member, *Concrete Canoe VIP* - New York, NY — Oct 2024 - May 2025

- Contributed to hands-on construction of the competition-grade concrete canoe, including shaping, and structural development.
- Developed prototype features for an iOS and Apple Watch app in Swift to collect accelerometer data and export CSV files for paddling analysis.

Experience

Teaching Assistant, *Intro to Engineering & Design, New York University* - New York, NY — Sep 2026 - Present

- Co-lead a laboratory section of 16 undergraduate students, guiding and assisting students with technical troubleshooting, prototyping, and applying the engineering design process.

Techs of Tomorrow, *Intrepid Museum (aviation and space museum)* - New York, NY — Summer 2024

- Developed hands-on projects with mentorship focused on aviation and aerospace concepts.
- Built experience with Code.org, Tinkercad, Scratch, and Cedero through design-based projects.

Tech360 Student & Tech Flex Leader, *America on Tech, Inc.* - New York, NY — Aug 2023 - May 2024

- Completed 100+ hours of technical coding in HTML, CSS, JavaScript, Repl.it, and Bootstrap.
- Presented the two final projects involving mock businesses and corresponding websites.
- Participated in weekly software engineering instruction and project-based learning workshops.

SKILLS

Hardware: ESP32, Raspberry Pi, FPGA, PCB Design, Soldering, Circuit Design/Analysis, Multimeter
Programming: Python, C++, Verilog, HTML/CSS
Engineering Tools: Altium Designer, Vivado, MATLAB, Fusion 360, 3D printing
Certificate: GWC - Intro to Data Science
Languages: Nepali (Native), English (Fluent), and Hindi (Intermediate)